

Sai Laxmi Tumu

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Profile

AI/ML Engineer with strong expertise in Generative AI, Large Language Models (LLMs), agentic AI systems, and Retrieval-Augmented Generation (RAG) architectures. Experienced in LLM fine-tuning, autonomous agent design, and scalable AI platform development using Python, Hugging Face Transformers, LangChain, and vector databases. Proven track record of deploying production-grade AI solutions on AWS and Google Cloud with a focus on performance, security, and responsible AI practices. Adept at collaborating with cross-functional teams to build intelligent systems that drive measurable business impact.

Skills

Programming Languages: Python, Java, SQL

GenAI & Machine Learning: LangChain, Hugging Face Transformers, TensorFlow, PyTorch, OpenAI API, Google Generative AI (Gemini), NLP, Prompt Engineering, Self-Supervised Learning, Generative AI Models, LLMs (GPT, LLaMA), LLM Fine-Tuning, Model Evaluation

Agents & Autonomous AI: Autonomous Agents, Multi-Agent Systems, Agentic AI Patterns, Planner-Executor Workflows, Decision-Making Agents, Tool-Augmented Agents, Safety-Constrained Agent Design

Vector Databases & Search: FAISS, RAG Implementations, Embedding Generation, Semantic Search, Context Window Optimization, Document Chunking Strategies

Cloud & DevOps: AWS, Google Cloud (GCP), Azure, Docker, Kubernetes, Helm, CI/CD Pipelines, Serverless Architectures

Data Engineering & Pipelines: Apache Airflow, PySpark, ETL Pipeline Development, Data Validation & Cleansing, BigQuery, Data Ingestion & Transformation

Backend & Frameworks: FastAPI, RESTful API Design, Microservices Architecture, Java Spring Boot, Hibernate, Spring Data JPA

Frontend & AI Interfaces: Streamlit, React, Jupyter Notebooks, Pandas

Monitoring, Logging & Observability: Stackdriver, Airflow Callbacks, Pipeline Status Logging, Model Monitoring & Metrics

MLOps, Governance & Compliance: MLOps, Model Deployment & Versioning, Data Privacy & Security, Responsible AI Practices, Model Risk Management

Experience

FIDELITY INVESTMENTS Jan 2024 - Present
AI ENGINEER

- Designed and deployed enterprise AI/ML solutions to enhance security, fraud prevention, and customer experience across financial platforms supporting **millions of users**.
- Built production-grade Generative AI and **Retrieval-Augmented Generation (RAG)** pipelines for call transcript summarization, customer insight extraction, and internal knowledge retrieval across research and advisory platforms.
- Implemented fraud detection and anomaly detection models that improved fraud identification accuracy by **30%+** and reduced false positives by **25%**, strengthening transaction security and risk controls.
- Developed scalable model training, deployment, and monitoring pipelines on **AWS and Google Cloud** using MLOps best practices for versioning, retraining, and performance tracking.
- Architected RAG systems using vector databases and semantic search to enable low-latency retrieval over large document repositories, improving advisor and operations productivity.
- Optimized feature engineering and inference pipelines, reducing model latency by **35%** and improving throughput for high-volume financial transaction workflows.
- Built AI-powered dashboards and reporting solutions to deliver real-time risk, compliance, and customer insights, enabling data-driven decisions across **millions of daily transactions**.

Cognizant Technology Aug 2020 - Aug 2022
ML Engineer

- Developed and deployed machine learning models for enterprise use cases including **predictive analytics, customer segmentation, and anomaly detection**, improving business KPIs by up to **25%** across digital transformation initiatives.
- Designed and implemented scalable data preprocessing and feature engineering pipelines using Python and SQL, reducing time to model training by **40%** and increasing data quality for downstream modeling.
- Built and productionized NLP solutions for unstructured data tasks such as sentiment analysis, entity extraction, and text classification, processing **millions of records per month** with near real-time performance.

- Collaborated with cross-functional teams to integrate ML models into client applications and dashboards, delivering actionable insights that accelerated decision-making for internal and customer-facing products.
- Implemented model monitoring and versioning processes using MLOps best practices, enabling automated retraining and performance tracking in cloud environments (AWS/GCP/Azure) for robust model governance.
- Optimized end-to-end model accuracy and latency through hyperparameter tuning, ensemble methods, and efficient inference pipelines, achieving production readiness with <1 second response times on key services.
- Produced technical documentation, model evaluation reports, and communicated results to business stakeholders to translate analytical insights into measurable outcomes for strategic initiatives.

Education

UNIVERSITY OF CINCINNATI, USA	Aug 2022 - Dec 2023
MS, INFORMATION TECHNOLOGY	GPA: 3.94/4.00
CMR INSTITUTE OF TECHNOLOGY, HYDERABAD, INDIA	Aug 2017 - Aug 2021
BACHELOR OF ENGINEERING, ELECTRONICS AND COMMUNICATION	GPA: 9.00/10

Projects

Smart Summary — AI-Powered Chrome Extension for Text Summarization

- Designed and built a production-grade AI-powered Chrome extension that extracts clean article content using content-script injection and generates executive-style summaries via a secure FastAPI backend and LLM integration.
- Implemented DOM extraction and content cleanup using Readability.js to remove ads, navigation, and layout noise, improving summarization quality and reducing token usage.
- Architected a scalable backend microservice with input validation, payload truncation (~20,000 characters), and structured JSON output, ensuring fast (~1–3s) and cost-efficient summarization.
- Designed the system to support hierarchical MapReduce-style chunked summarization for long-form research papers and technical documents

Information Retrieval System with RAG Architecture

- Developed a production-ready RAG system integrating Google's Gemini LLM, LangChain framework, FAISS vector database, and custom prompt engineering to enable accurate document-grounded responses
- Implemented advanced document processing pipeline with chunking strategies, embedding generation, and context window optimization for multi-format documents (PDF, DOCX, TXT)
- Created an interactive Streamlit interface with natural language querying capabilities, achieving 90% query relevance while optimizing token usage and reducing inference costs

Text-to-Image Generation Application

- Architected a text-to-image generation system leveraging Hugging Face's ART_v1.0 model and inference API, enabling non-technical users to create high-quality images from text descriptions
- Implemented advanced image processing pipeline with PIL for post-generation enhancements, metadata management, and local storage optimization
- Built responsive Streamlit web interface with gallery functionality, achieving 40% faster image generation through optimized request handling and caching strategies

MCQ Generator with LLMs

- Engineered an educational content generation system using LangChain, OpenAI APIs, and advanced prompt engineering to transform complex topics into pedagogically sound multiple-choice questions
- Developed sophisticated evaluation mechanisms for quality control, ensuring high accuracy in content relevance and answer correctness across diverse subject domains
- Implemented comprehensive analytics dashboard tracking token usage, response latency, and cost metrics to optimize API consumption and improve system performance

Certifications & Publications

- **AI Agents Fundamentals** | Hugging Face — May 2025
Skills: Agentic AI, Autonomous Agents, Multi-Agent Systems, Tool-Augmented Agents
- **Finetuning Large Language Models** | DeepLearning.AI — Mar 2025
Skills: LLM Fine-Tuning, PyTorch, Transformers, Deep Learning
- **Data-Driven Graduate Certificate (Cybersecurity)** | University of Cincinnati — Jan 2023
Skills: Cybersecurity, Data Security, Forensic Analysis
- **Advance Your Skills in Deep Learning and Neural Networks** | LinkedIn Learning — Dec 2022

